Supplement: parameter-sensitivity methods

Supplement: parameter-sensitivity methods I

This supplement documents how the sensitivity grid of sec. 13.13 is generated and — importantly for a reader trying to reconcile numbers across studies — where its two sweeps use *different* seeding and trial budgets. It answers two questions the main section leaves implicit: exactly which seed drives each cell (so any cell is independently reproducible), and why the cross-study summary's sensitivity row is not directly comparable, at matched trial counts, to the standalone heatmap.

The sensitivity sweep is implemented in `fedference.experiments.run_belief_sharing_sensitivity` and `fedference.experiments.run_hierarchical_sensitivity`. Each function accepts a tuple of acuity values and a tuple of colony sizes. In the **belief-sharing** sweep every (acuity, colony-size) cell averages n_{trials} independent trials, each seeded via a deterministic formula:

Experimental protocol for grid sensitivity I

$$\mathsf{seed}_{\mathsf{cell}} = \mathsf{seed}_{\mathsf{base}} + i \cdot 10^5 + j \cdot 10^3 + t \tag{1}$$

The deterministic seed rule eq. 1 makes every grid cell and trial independently reproducible from the base seed.

where i indexes acuity, j indexes colony size, and t indexes the trial within a cell. For the belief-sharing sweep this guarantees that:

Experimental protocol for grid sensitivity I

1. no two cells share a trial seed (no correlation between cells);
2. re-running with the same `seed_base` is bit-identical (reproducibility);
3. different `seed_base` values produce independent replicates (robustness checking).

Experimental protocol for grid sensitivity I

The **hierarchical** sweep uses a simpler protocol: every cell calls `run_hierarchical_world` once with the same base seed (its internal trials are seeded by that run), so hierarchical cells share the base seed rather than the per-cell formula above.

Grid parameters for acuity and colony size I

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Parameter	Values
Sensor acuity κ	$\{0.40, 0.55, 0.70, 0.85, 0.95\}$
Colony size n	$\{2, 4, 6, 8, 10\}$
Trials per cell	20
Base seed	0

Grid parameters for acuity and colony size I

The $5 \times 5 = 25$ cells per system are run with `seed_base = 0` by default; `generate_sensitivity_heatmap` accepts a `seed` argument to override this.

Belief-sharing condition in the sensitivity grid I

Each trial in the belief-sharing sweep:

Belief-sharing condition in the sensitivity grid I

1. Draws a random true state and one noisy observation per agent (same protocol as `run_belief_sharing`).
2. Runs one belief-sharing round with `communicate=True` (communicating) and `communicate=False` (isolated).
3. Records `mean_accuracy` for each condition.
4. The cell value is the average over `n_trials` of this gap.

Hierarchical POMDP condition in the sensitivity grid I

Each cell in the hierarchical sweep calls `run_hierarchical_worl d` once with the cell's acuity and colony size, passing the constant base seed (not the per-cell formula, which applies only to the belief-sharing sweep). The returned `location_accuracy_gap` (hierarchical minus flat) becomes the cell value.

`generate_sensitivity_heatmap` assembles the two grids into a 1×2 matplotlib `imshow` figure with RdYlGn colormap, symmetric bounds at $\pm\max(|\text{gap}|)$, per-cell numeric annotations, and a per-panel colorbar labeled "Accuracy gap (hierarchical/comm. − baseline)". The figure is written to `output/figures/sensitivity_heatmap.png`.

`generate_cross_study_summary` runs a 128-seed ($n_{\text{seeds}} = 128$) ensemble over Studies 1–9 and reports the mean $\pm$ 95 % bootstrap CI of the key federation-benefit metric for each study. The metric definitions are:

The robustness row uses 40 matched trials per seed and rate; the trial-level observations are reduced within seed before the cross-study summary is formed. This preserves the seed as the independent Monte Carlo unit.

Cross-study summary construction I

The Study 8 row below uses 3 trials per cell — smaller than the full-resolution 20-trial `Trials per cell` grid documented above for the standalone sensitivity heatmap figure, a deliberate runtime budget for the per-seed cross-study loop rather than an oversight — so the two are not directly comparable at matched trial counts.

Cross-study summary construction I

Study	Metric
1 — Belief sharing	Accuracy gain: communicating − isolated
2 — Language acquisition	KL reduction: initial − final
3 — Emergence (BMR)	ΔF for redundant pruning
4 — Robustness sweep	Accuracy gain: pooled display robust method − naive at worst contamination rate
5 — Moving world (EFE)	Accuracy gain: EFE-guided − isolated
6 — Hierarchical POMDP (2-level)	Location accuracy gap: hierarchical − flat
7 — 3-level POMDP	Location accuracy gap: 3-level − flat
8 — Parameter sensitivity	Mean accuracy gap across the sensitivity grid
9 — Parameter recovery	R^2 for acuity identifiability

Cross-study summary construction I

Bootstrap CIs use 5000 resamples (default `n_boot` in `fedference.statistics.bootstrap_ci`).